

Equations with Brackets

AQA · KS3 Year 8

Answer each question in your workbook. Show your working.

Method

Expand every bracket fully Collect like terms on each side Move the smaller letter term across
Collect numbers, then divide Check in the original equation

Common mistakes

- $3(x + 2) = 3x + 6$, not $3x + 2$.
- A negative outside flips both signs inside.
- Do not divide before expanding when other terms are present.
- Watch signs when subtracting a bracket, e.g. $-2(x - 4) = -2x + 8$.

Worked example

Solve $3(x + 2) = 18$.

Expand: $3x + 6 = 18$. Subtract 6: $3x = 12$.
Divide by 3: $x = 4$.

Check: $3(4 + 2) = 18$

Visual aid

Visual Example

$$2x + 3 = 11$$

$$\text{-3 both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



Area model: a rectangle split into 3 lots of $(x + 2)$. The total area $3x + 6$ equals 18, so each $x = 4$.

Questions

Answer each question in your workbook. Show your working.

Q1 of 8

[2] ● ○ ○ ○ ○

Solve: a) $2(x + 3) = 14$ b) $5(x - 1) = 20$

Q2 of 8

[2] ● ○ ○ ○ ○

Solve: a) $3(x + 4) = 18$ b) $4(2x - 1) = 28$

Q3 of 8

[3] ● ● ○ ○ ○

Solve: a) $2(3x + 1) = 26$ b) $5(2x - 3) = 35$ c) $3(x + 5) = 12$

Q4 of 8

[3] ● ● ○ ○ ○

Solve: a) $2(x + 4) = x + 11$ b) $4(x - 1) = 2(x + 3)$ c) $3(2x + 1) = 4x + 11$

Q5 of 8

[3] ● ● ● ○ ○

Solve: a) $2(x + 1) + 3(x - 2) = 11$ b) $5(x - 1) - 2(x - 4) = 18$ c) $4(x + 2) - 3(x - 1) = 15$

Q6 of 8

[3] ● ● ● ○ ○

Rectangle length $(x+5)$, width 3, perimeter 28 cm. a) Equation. b) Find x . c) Area.

Q7 of 8

[4] ● ● ● ● ○

Regular pentagon, side $(2x-1)$ cm, perimeter 45 cm. a) Equation. b) Find x . c) Side length.

Q8 of 8

[5] ● ● ● ● ●

a) Solve $3(2x - 1) = 2(x + 5) + 3$. b) Show $4(x + 2) = 2(2x - 1)$ has no solution and explain.